PROJECT DEMO PLANNING

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Introduction

The Rising Waters project demonstration is planned to showcase the complete functionality of the AI-powered Flood Prediction System. The demonstration explains the project workflow, machine learning model, user interface, prediction process, and deployment.

Demonstration Objectives

- Explain the project overview.
- Demonstrate the Streamlit web application.
- Show flood prediction using sample inputs.
- Explain the Machine Learning model.
- Present the project outcomes and benefits.

Demonstration Plan

Step 1 – Project Introduction

Introduce the Rising Waters project, its objectives, and the problem it solves.

Step 2 – Technology Overview

Explain the technologies used:

- Python
- Streamlit
- Pandas

- NumPy
- Scikit-learn
- XGBoost
- GitHub

Step 3 – Dataset Explanation

Describe the rainfall and environmental dataset used for training the Machine Learning model.

Step 4 – Application Demonstration

Open the Streamlit application and explain the dashboard.

Step 5 – Prediction Process

Enter rainfall, temperature, and humidity values.

Click the **Predict Flood Risk** button.

Display the prediction result.

Step 6 – Result Explanation

Explain how the Machine Learning model predicts flood risk and how the result helps users.

Step 7 – Deployment

Demonstrate the deployed Streamlit application and GitHub repository.

Required Resources

- Laptop/Desktop
- Internet Connection
- Streamlit Application
- GitHub Repository
- Project Documentation

Expected Outcome

- Successful project demonstration.

- Accurate flood prediction.
- Smooth application performance.
- Better understanding of the project workflow.

Benefits

- Easy to explain.
- Professional presentation.
- Clear demonstration process.
- Improved project understanding.

Conclusion

The project demonstration plan ensures that all important features of the Rising Waters application are presented in a structured and professional manner. It highlights the working of the Machine Learning model and the overall functionality of the system.